

Elective without Specilaization

Programme	B. Sc. Computer Science				
Course Title	Microprocessor and its Applications				
Type of Course	Elective				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	-				
Course Summary	This course provides a comprehensive understanding of microprocessors with a specific focus on Intel's 8085 & 8086 architectures. The course delves into the key principles, features, and programming techniques associated with 8086. The feature comparison of advanced processors gives an overview of developments in Microprocessor technology.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Recognise the purpose, characteristics, and architecture of the 8085 and 8086.	U	C	Instructor-created exams / Quiz
CO2	Identify the addressing modes and comprehend how the 8086 instructions work.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Illustrate simple assembly language programs.	Ap	P	Programming Assignment / Observation of Practical Skills
CO4	Identify the functions of peripheral integrated circuits (ICs) and how interrupts are handled in the 8086.	U	C	Instructor-created exams / Seminars
CO5	Describe the characteristics of advanced microprocessors.	U	C	Instructor-created exams / Home Assignments
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks (70)
I	Introduction to Microprocessors		8	15
	1	Basic Architecture of a Computer System	1	
	2	Advances in Semiconductor Technology, Evolution of Microprocessors	2	
	3	Overview of Microprocessors vs Microcontrollers, Computer Languages: High Level, Machine Language, Assembly Language	1	
	4	8085 Microprocessor (Architecture and Pin diagram)	4	
II	Basics of 8086 Architecture		9	20
	5	Features of an n-bit microprocessor, 8086 Architecture (Block diagram, Register Configuration, Address Translation)	5	
	6	8086 Pin Configuration	2	
	7	Minimum and Maximum Mode Configuration	2	
III	8086 Programming		23	20
	8	8086 Addressing Modes	2	
	9	8086 Instruction Set: Data Transfer and Arithmetic Instruction	4	
	10	8086 Instruction Set: Branch and Loop Instructions	3	
	11	8086 Instruction Set: Sting Instructions	2	
	12	8086 Instruction Set: Processor Control Instructions	1	

	13	Assembler Directives	1	
	14	Sample Programs: 1) Assembly Program to find the sum of n numbers given 2) Assembly Program to perform division using repeated subtraction 3) Assembly Program to multiply two 16 bit numbers 4) Assembly Program to find the largest of n numbers given 5) Assembly Program to perform linear search in a set of numbers given. Also find the number occurrence of the searching element. 6) Assembly Program to perform comparison of two strings.	6	
	15	8086 Interrupts and Interrupt Service Routines	2	
	16	Procedures and Macros	2	
IV	Advanced Microprocessors (Study of Architecture and Pin diagram not needed)		8	15
	17	Features of Intel 80186 & 80286	2	
	18	Features of Intel 80386 & 80486	2	
	19	Features of Pentium Processors	1	
	20	Features of Multi Core Processors	1	
	21	Features of i series Processors	1	
	22	Features of Mobile Processors	1	
V	Open Ended Module: 8086 Interfacing		12	
	1	Introduction to peripheral Interfacing Include Case studies of any 3 interfacing ICs like: 1. Programmable Peripheral Interface (8255) 2. Programmable DMA Controller (8257) 3. Programmable Interrupt Controller (8259) 4. Programmable Interval Timer (8253) 5. Interfacing output displays (8212) 6. Programmable communication interface (8251A)	12	

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	3	-	-	-	-						
CO 2	2	2	-	1	-	-						
CO 3	-	-	-	2	-	-						
CO 4	2	2	-	-	-	-						
CO 5	1	1	-	-	-	1						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment/Seminar	Programming Assignments	End Semester Examinations
CO 1	✓			✓
CO 2	✓		✓	✓
CO 3	✓		✓	✓
CO 4	✓	✓		
CO 5		✓		✓